

GFIS Mid-Semester Presentation Script

Opening

Good morning respected evaluator and faculty members. My dissertation project is titled Green Fuel Intelligence System, or GFIS. The work focuses on methane yield prediction, soft sensing, and digital-twin-ready simulation for anaerobic digestion using a physics-guided hybrid AI approach.

The abstract presentation was accepted with an Excellent remark. The feedback encouraged strengthening the work as a Physics-Informed Machine Learning framework with digital-twin-ready simulation capability. I have used that feedback to refine the mid-semester work from a prediction application into a research prototype with model evaluation, process constraints, soft sensing, and scenario simulation.

Problem Statement

Anaerobic digestion is a nonlinear biological and chemical process. Methane yield is affected by temperature, pH, organic loading rate, hydraulic retention time, total solids, volatile solids, C/N ratio, moisture, and accumulated reactor state. If the model only predicts methane yield without process context, it can become a black-box tool and may not support operator decisions.

GFIS therefore tries to answer three questions:

1. What methane yield is expected under the current operating condition?
2. Is the operating condition stable, warning, or critical?
3. Does the prediction remain within a physically feasible methane bound?

System Design

The system has four main layers.

First, the data layer generates and processes a synthetic/literature-inspired anaerobic digestion dataset. It handles process variables, missing values, normalization, lag features, rolling features, and time windows for sequential modelling.

Second, the model layer includes an XGBoost tabular model, a PyTorch LSTM branch, an ensemble prediction path, and a physics-guided ensemble check. The physics layer applies a volatile-solids-based methane upper bound and records violation counts.

Third, the soft-sensor layer estimates the VFA/ALK ratio and produces a stability label: Stable, Warning, or Critical.

Fourth, the interface layer includes FastAPI endpoints, Streamlit dashboard, HTML Level 2 portal, and an industrial simulation/control-room interface.

Current Evidence

The current evaluation is preliminary and based on a synthetic/literature-inspired anaerobic digestion dataset. The tabular XGBoost model currently performs best, with R^2 of 0.7556, RMSE of 6.7933, and MAE of 5.5517. The physics-guided ensemble reports zero physics-bound violations in the current evaluation.

The LSTM branch has weaker performance at this stage, with R^2 of 0.2942. I have retained it because the final dissertation direction includes real sequential plant data, where temporal dependencies and retention-time effects can become more important.

Scenario Experiments

Beyond one-step prediction, GFIS includes 48-hour controlled scenario experiments. These include nominal mesophilic operation, organic overload, pH acidification shock, low-temperature disturbance, reduced HRT washout risk, high-solids feedstock, and corrective recovery operation.

The important point is that the system is not only giving a number. It is showing how methane yield, VFA/ALK ratio, and stability behave under process disturbances. For example, the organic overload scenario activates the warning pathway through the soft sensor.

Digital Twin Positioning

I am presenting the current system as a digital-twin-ready simulation dashboard, not as a complete industrial digital twin. A full industrial digital twin would require continuous connection with a physical biogas plant, IoT or SCADA data stream, and live calibration against plant behavior.

The current work builds the foundation: data pipeline, models, physics constraints, soft sensing, simulation, APIs, dashboard, and deployment structure.

Demonstration Flow

For the live demonstration, I will show:

1. The GFIS portal and Level 2 workbench.
2. The industrial simulation/control-room interface.
3. A prediction or simulation run.
4. Stability output and VFA/ALK estimate.
5. Evidence files such as evaluation report, scenario experiments, and dataset.

If local services are running, the expected interfaces are:

- FastAPI backend: `http://127.0.0.1:8010`
- HTML portal/simulation: `http://127.0.0.1:8520`
- Streamlit dashboard: `http://127.0.0.1:8510`

Limitations

The present version uses synthetic/literature-inspired data, so I will not claim real-plant validation yet. The system is also not continuously connected to IoT/SCADA. The final phase will focus on improving data quality, adding stronger explainability, testing uncertainty, and preparing dissertation-grade validation.

Closing

In summary, GFIS has progressed from accepted abstract to a working mid-semester prototype. The contribution is a physics-guided hybrid AI framework that combines methane prediction, soft sensing, stability classification, simulation, and deployment-ready interfaces for anaerobic digestion research.